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Nowcasting: Practical Results with Theoretical Flaws

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Outline

- Motivation
- Nowcasting using DFMs
- EU-GDP example
- Simulation Study
- Thoughts going forward

Acknowledgements

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Thanks to Sogeti for subcontracting us to fulfil this research.

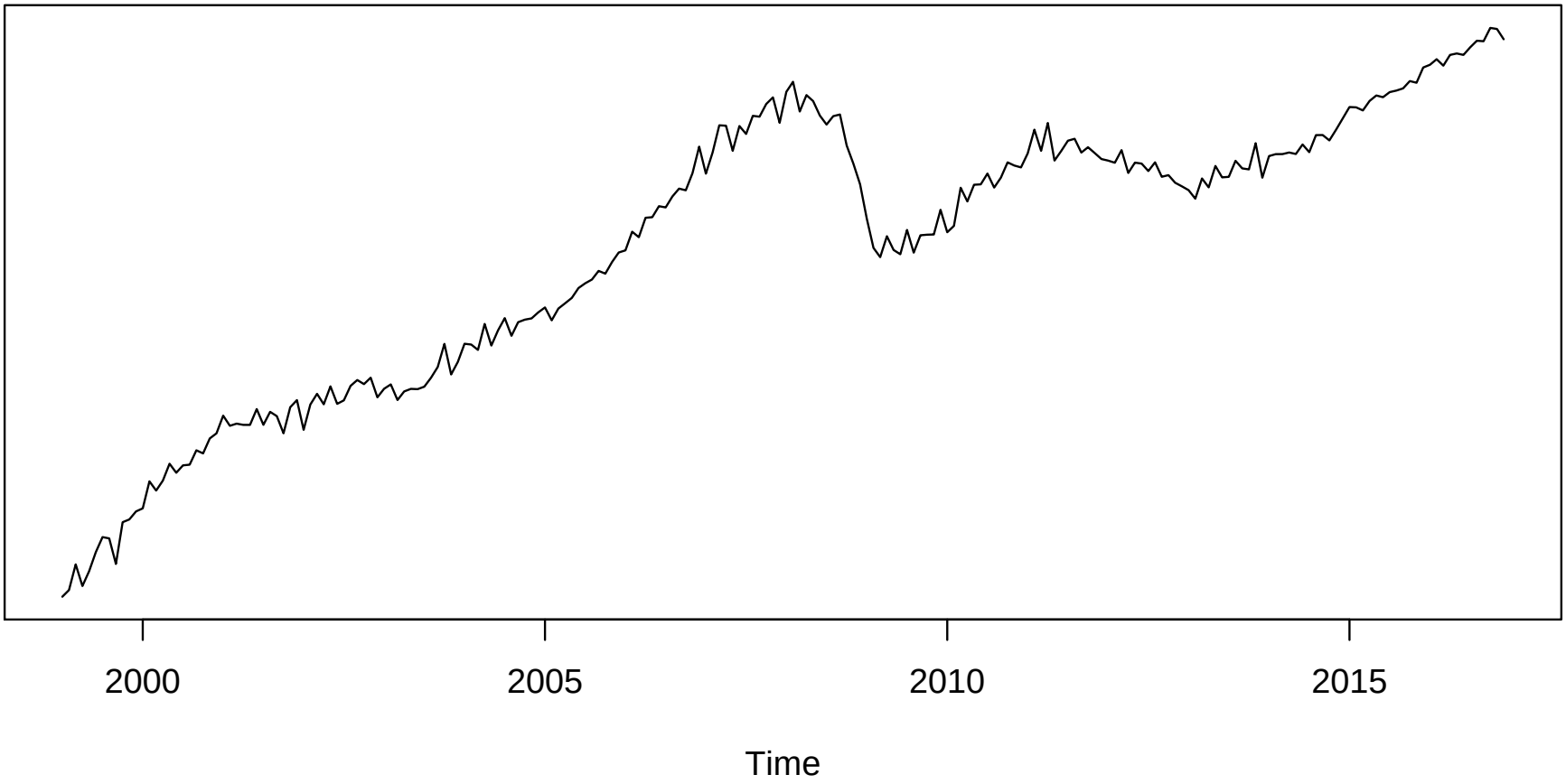
The majority of this work was completed by Ben Norwood, a recent PhD student.

The views expressed are my own.

Motivation

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We want to give a prediction of GDP based on latest available data (arriving at different frequencies).



Nowcasting using DFMs

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General Dynamic Factor Model (DFM):

$$Y_t = \lambda(L)F_t + v_t \qquad v_t \sim N(0, V)$$
$$F_t = \psi(L)F_{t-1} + w_t \qquad w_t \sim N(0, W)$$

Without lag polynomials:

$$Y_t = AF_t + v_t \qquad v_t \sim N(0, V)$$
$$F_t = F_{t-1} + w_t \qquad w_t \sim N(0, W)$$

Need to estimate the $T \times p$ factor loading matrix A and correlation matrices V and W.

Identifiability

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$$A = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ a_{2,1} & 1 & 0 & \cdots & 0 \\ a_{3,1} & a_{3,2} & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{p,1} & a_{p,2} & a_{p,3} & \cdots & 1 \\ a_{p+1,1} & a_{p+1,2} & a_{p+1,3} & \cdots & a_{p+1,p} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m+1,1} & a_{m+1,2} & a_{m+1,4} & \cdots & a_{m+1,p} \end{bmatrix}, W = \begin{bmatrix} w_1 & 0 & 0 & \cdots & 0 \\ 0 & w_2 & 0 & \cdots & 0 \\ 0 & 0 & w_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & w_p \end{bmatrix}$$

How do we estimate p and m?

Previous Work

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Choosing *p*, number of factors:

- Domain expertise
- Scree Plots - Cattell (1966), Onatski (2009)
- Information Criteria - Bai & Ng (2002)

See Hoyle & Duvall (2004) and Eurostat Handbook on Rapid Estimates (2017).

Choosing *m*, number of relevant series for prediction:

- Domain expertise
- LASSO / Elastic Net - Bai & Ng (2008)
- Marginal predictive gain - Runstler (2016)

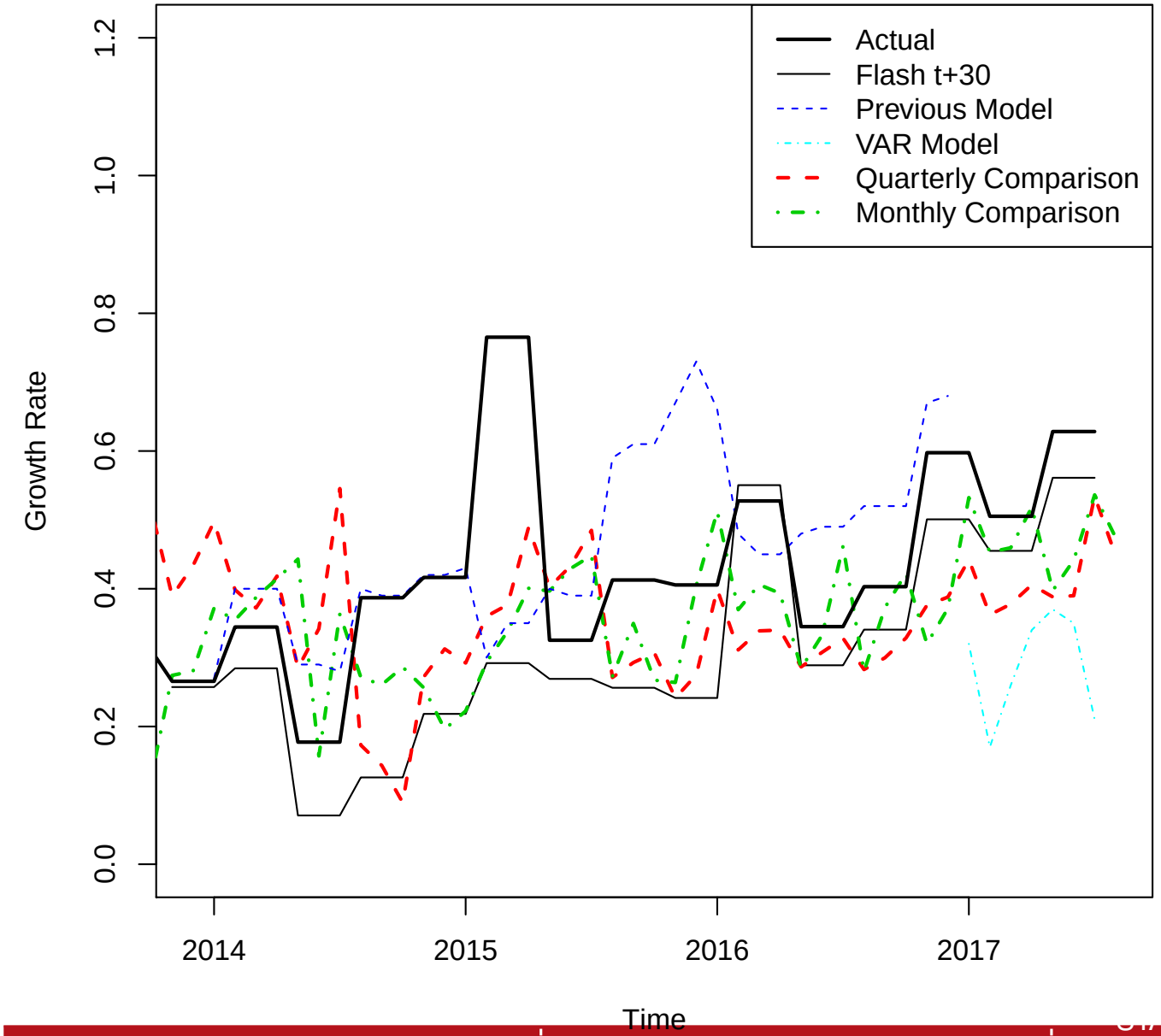
Our contribution

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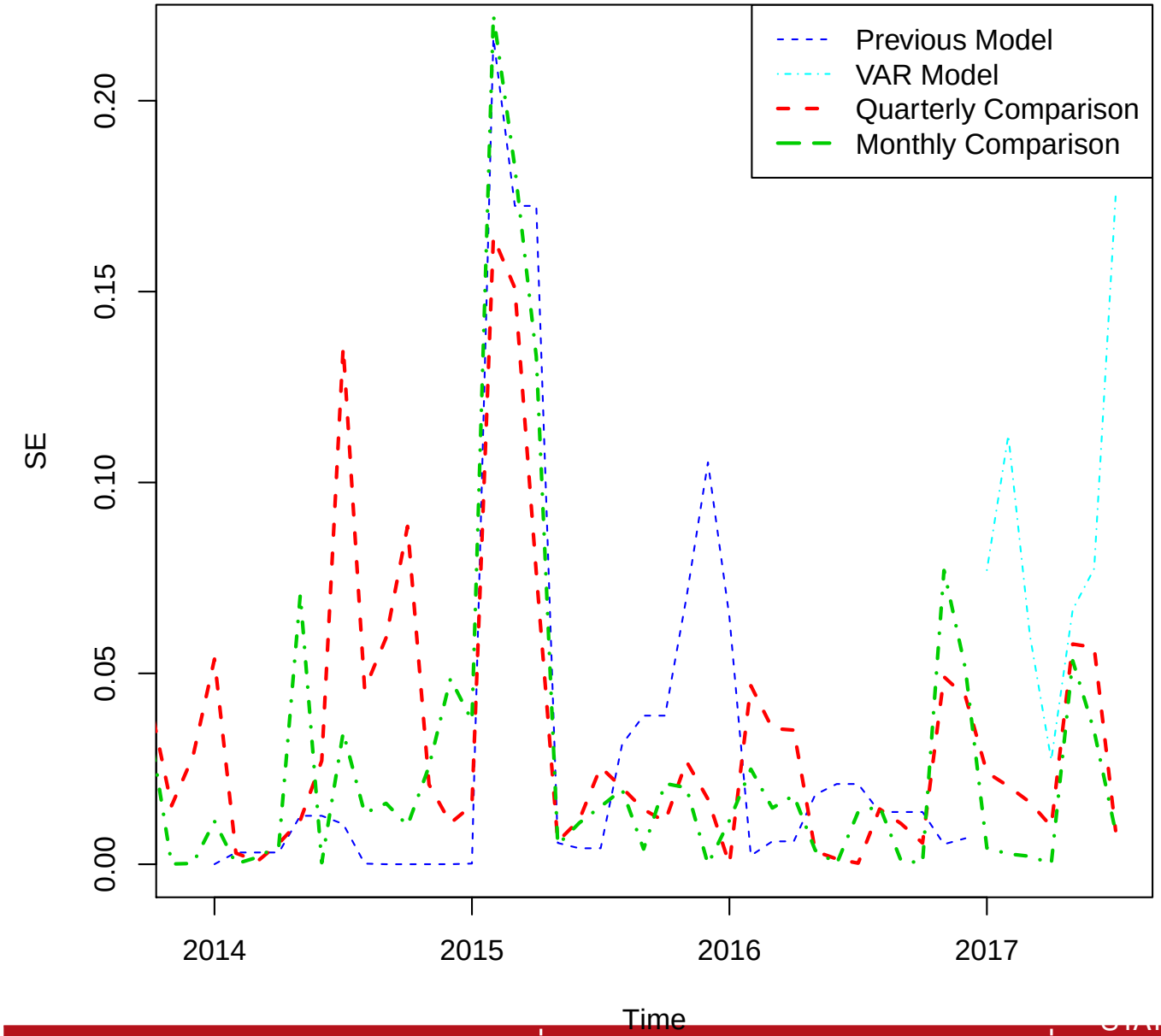
- Simultaneous estimation of p and m
- Dynamic estimation of p and m across time

We fit our DFM using the `dlm` package in `R`.

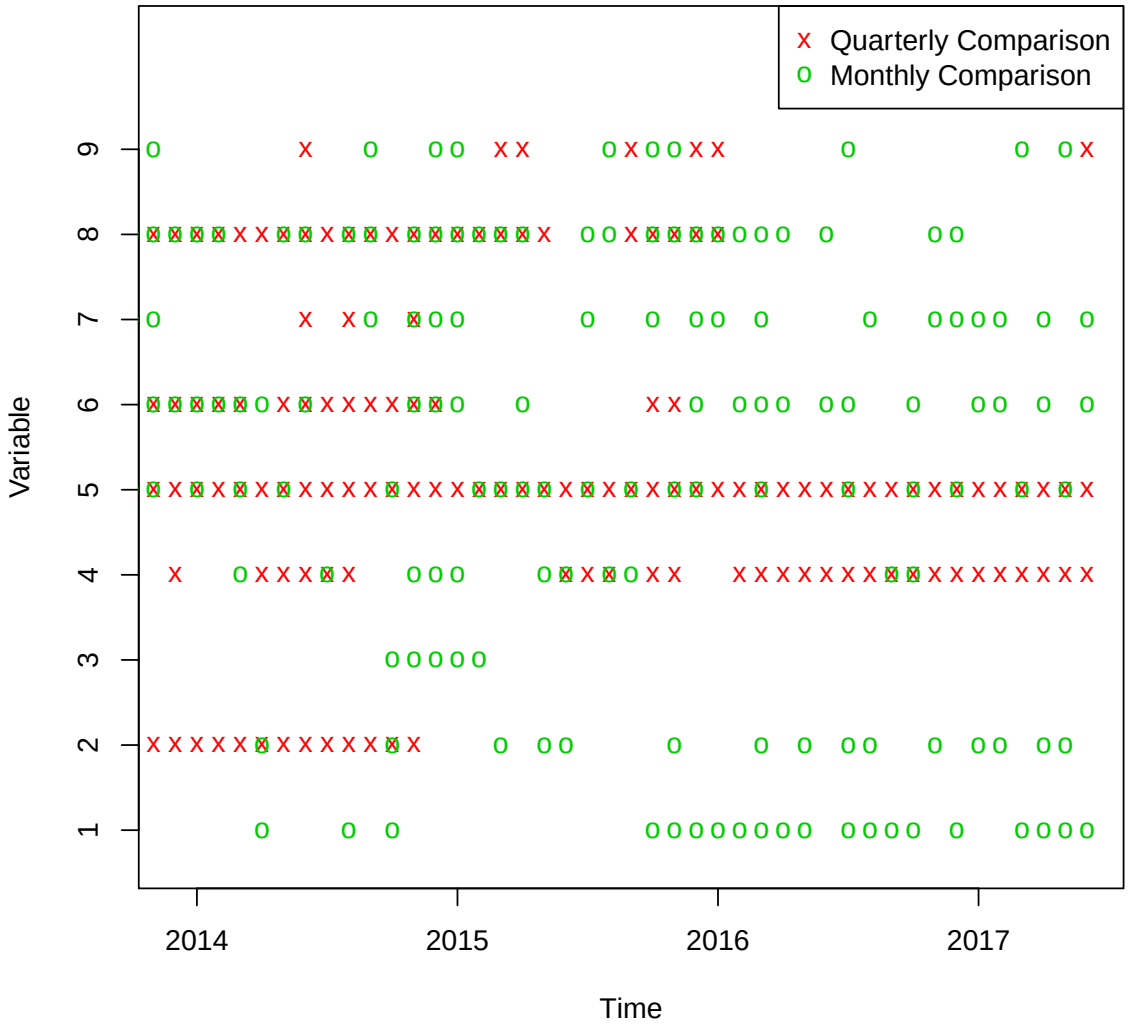
Results - Growth Rate



Results - Squared Error

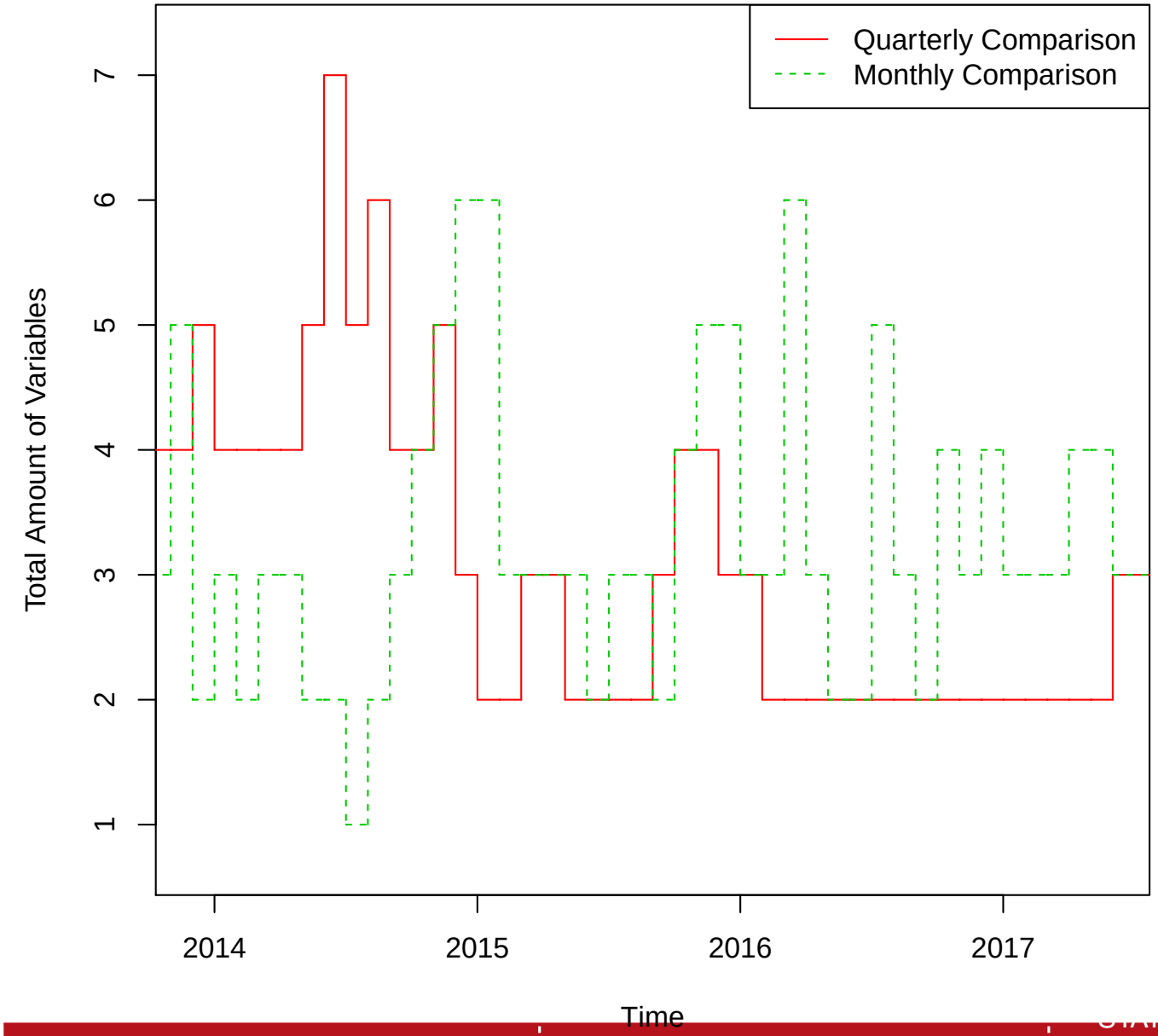


Results - Variables

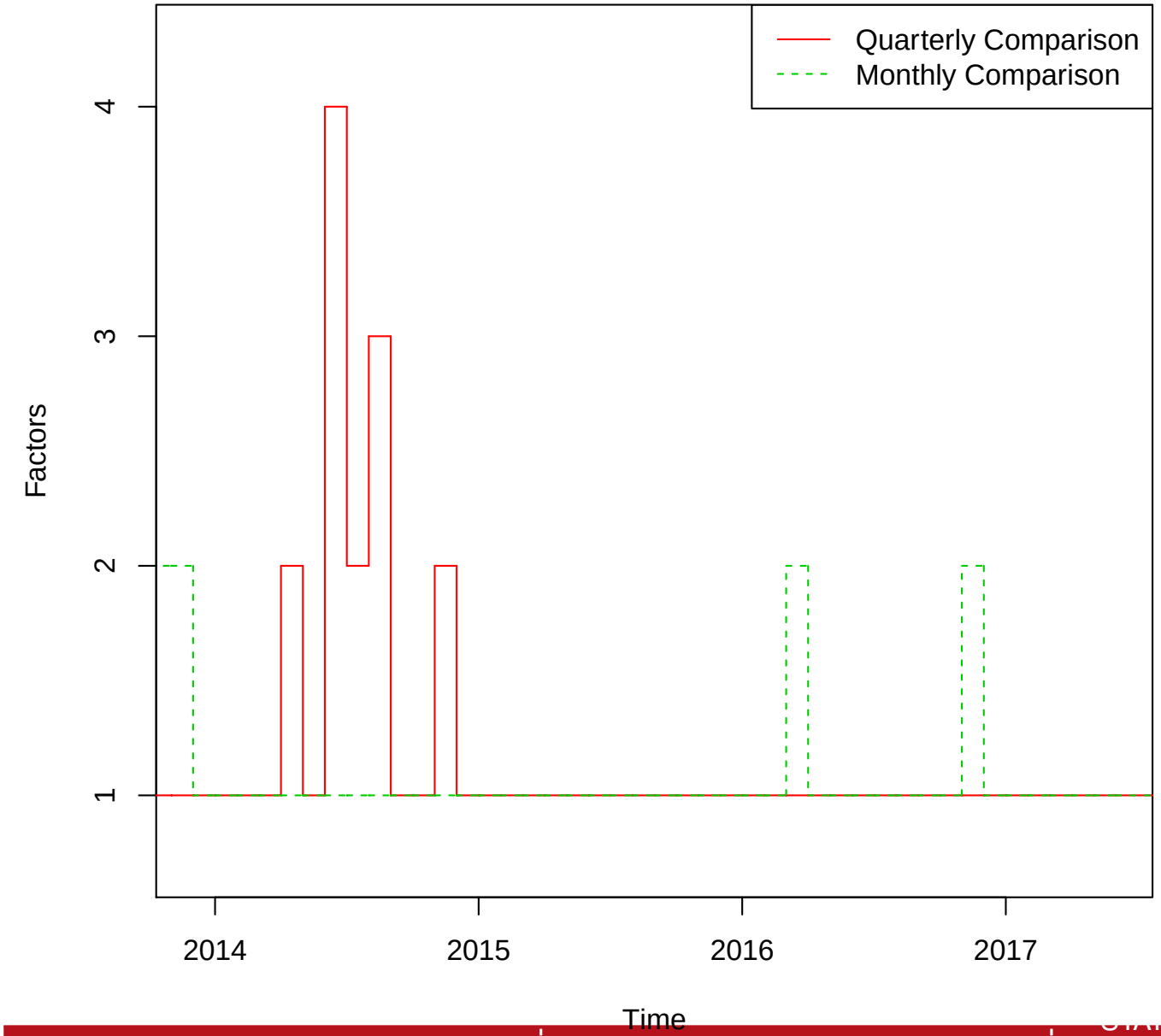


9- Retail trade
8- Industrial order book
7- Industrial Export OB
6- Construction Confience
5- Household Survey
4- Household 12 months
3- Economic sentiment
2- Industrial production
1- Euro/Dollar Exchange

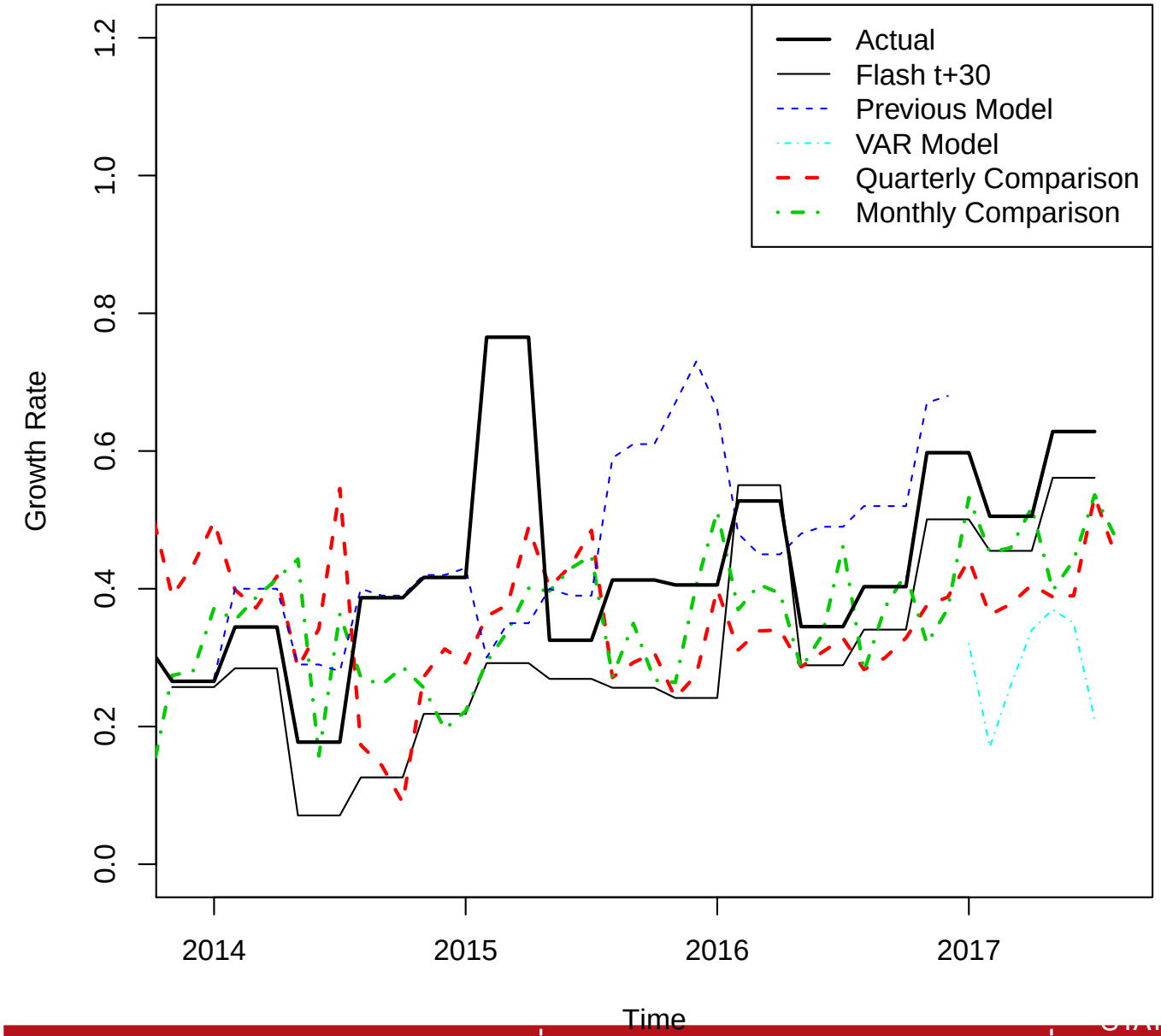
Results - Variables



Results - Variables



Results - Growth Rate



Simulation setup

- Twice as many variables as factors $p = 2m$
- Each variable contributes equally
- Each factor contributes equally

$$\text{Model 1: } p = 1, m = 3, A = \begin{bmatrix} 1 & 0.5 & -0.5 \end{bmatrix}^T,$$

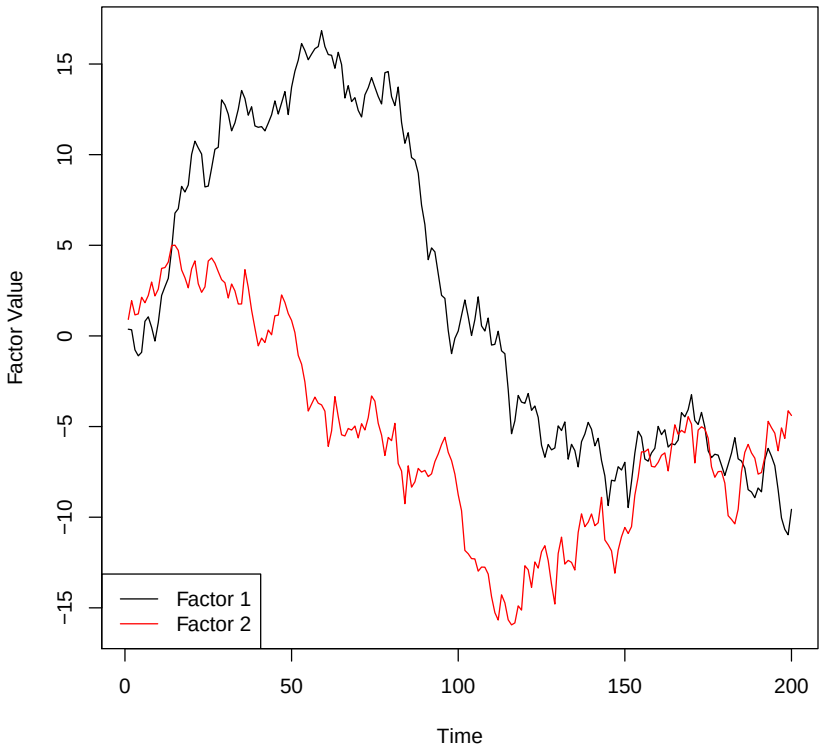
$$\text{Model 2: } p = 2, m = 5, A = \begin{bmatrix} 1 & 0.8 & -0.6 & 0.4 & -0.2 \\ 1 & 0.2 & 0.4 & 0.6 & 0.8 \end{bmatrix}^T,$$

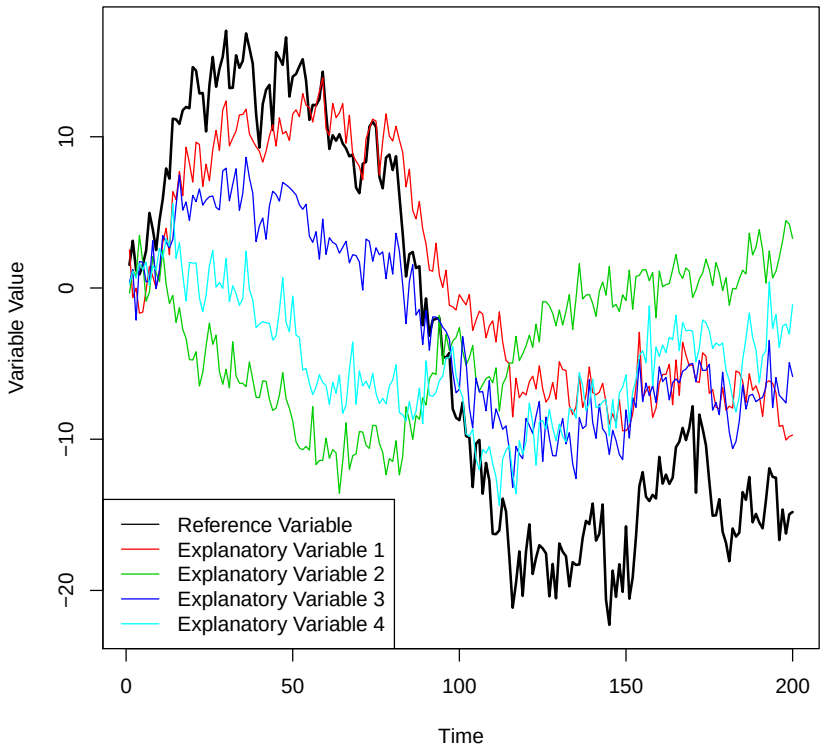
$$\text{Model 3: } p = 2, m = 5, A = \begin{bmatrix} 1 & 0.7 & -0.1 & 0.9 & -0.3 \\ 1 & 0.3 & 0.9 & 0.1 & 0.7 \end{bmatrix}^T,$$

$$\text{Model 4: } p = 2, m = 5, A = \begin{bmatrix} 1 & 0.66 & 0.1 & 0.2 & -0.285 \\ 1 & 0.33 & -0.4 & 0.3 & 0.215 \end{bmatrix}^T.$$

Independent series are variety of WN, AR, MA, ARMA.

Simulation


Factor Value
15
10
5
0
−5
−10
−15
0
50
100
150
200
Time
Factor 1
Factor 2


Variable Value
10
0
−10
−20
0
50
100
150
200
Time
Reference Variable
Explanatory Variable 1
Explanatory Variable 2
Explanatory Variable 3
Explanatory Variable 4

Results - Not-time varying

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Dep Model	Ind Model	Ind (n=) 100	Ind (n=) 200	Ind (n=) 500	Dep (n=) 100	Dep (n=) 200	Dep (n=) 500	Factors (n=) 100	Factors (n=) 200	Factors (n=) 500
1	1	0.78	0.81	0.83	0.24	0.20	0.26	0.91	0.90	0.95
1	2	0.73	0.75	0.83	0.25	0.25	0.25	0.90	0.92	0.95
1	3	0.75	0.86	0.78	0.27	0.23	0.24	0.86	0.95	0.94
1	4	0.76	0.72	0.81	0.00	0.00	0.00	0.91	0.90	0.93
2	1	0.80	0.84	0.75	0.02	0.04	0.04	0.07	0.09	0.10
2	2	0.75	0.82	0.74	0.02	0.02	0.01	0.10	0.04	0.07
2	3	0.80	0.78	0.74	0.02	0.04	0.02	0.10	0.08	0.07
2	4	0.77	0.82	0.75	0.00	0.00	0.00	0.08	0.05	0.09
3	1	0.83	0.88	0.86	0.01	0.03	0.03	0.08	0.10	0.03
3	2	0.76	0.83	0.80	0.02	0.03	0.02	0.12	0.08	0.08
3	3	0.85	0.81	0.80	0.02	0.03	0.05	0.06	0.09	0.09
3	4	0.80	0.84	0.79	0.00	0.00	0.00	0.09	0.10	0.06
4	1	0.81	0.80	0.93	0.04	0.05	0.02	0.11	0.16	0.04
4	2	0.84	0.71	0.85	0.04	0.07	0.01	0.11	0.13	0.06
4	3	0.84	0.75	0.81	0.04	0.07	0.01	0.11	0.17	0.08
4	4	0.83	0.71	0.83	0.00	0.00	0.00	0.12	0.14	0.05

What is going on?

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Could it be

- the information criterion chosen?
- we don't have enough data? (c.240 in GDP example)
- the simulation models chosen?
- the factor and variables relationships?
- that joint estimation doesn't produce consistent results?

Note that the DFM literature is also finding similar results.

Larger Simulation

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- Wide range of decision metrics for m and p
- Wide range of simulated models
- Used both forwards and backwards selection for m and p

Broadly the results are the same.

Summary

- DFM models practically work well at Nowcasting
- Time-varying estimation of factors and variables produces a reduction in MSE

However

- DFM models do not recover the true underlying structure reliably
- But they are good at removing independent variables
  All models are wrong, but some models are useful - Box

Is it useful if it isn't consistent?